

# FINAL REPORT

## Gym Equipment Management System

(CRUD Application)

## Student Details

| field             | Details                                     |
|-------------------|---------------------------------------------|
| Student Name      | Amandi M.D.S.A.                             |
| Student ID        | IT25101654                                  |
| Module            | Object Oriented Programming                 |
| Project Title     | Gym Equipment Management System             |
| Database          | MySQL                                       |
| Technologies used | Java,JSP,Servlet,JDBC,MySQL,HTML,CSS,GitHub |

## Introduction

The Gym Equipment Management System is a web-based CRUD application developed to manage gym equipment efficiently. The system allows administrators to add, update, delete, search, and manage gym equipment records while customers can view available equipment through a separate customer dashboard.

This system was developed using Java, JSP, Servlets, JDBC, MySQL, HTML, and CSS following Object-Oriented Programming concepts and DAO architecture.

## Objectives

The main objectives of this project are:

- To develop a complete CRUD system for gym equipment management.
- To implement Object-Oriented Programming concepts.
- To create separate admin and customer dashboards.
- To connect the system with a MySQL database.
- To implement file handling for activity logging.
- To design a professional and user-friendly interface.

# Technologies Used

| Technology    | Purpose               |
|---------------|-----------------------|
| Java          | Backend development   |
| JSP           | Dynamic web pages     |
| Servlets      | Request handling      |
| JDBC          | Database connectivity |
| MySQL         | Database management   |
| HTML/CSS      | Frontend UI design    |
| GitHub        | Version control       |
| Apache Tomcat | Web server            |

## System Features

### Admin Features

- Admin Login Authentication
- Add Equipment
- Update Equipment
- Delete Equipment
- Search Equipment
- Dashboard Overview
- View Total Equipment Count
- View Available Equipment Count
- View Broken Equipment Count
- Logout Functionality

### Customer Features

- View Equipment
- Search Equipment
- Pagination (Next / Previous)
- About Page Access

# CRUD Operations

| Operation | Description                |
|-----------|----------------------------|
| Create    | Add new gym equipment      |
| Read      | View equipment records     |
| Update    | Edit equipment information |
| Delete    | Remove equipment records   |

## Database Design

### Database Name

gymdb

### Table Name

equipment

### Table Structure

| Column Name      | Data Type |
|------------------|-----------|
| id               | int       |
| name             | varchar   |
| Type             | varchar   |
| quantity         | int       |
| condition_status | Varchar   |

# Object-Oriented Programming Concepts

## Encapsulation

The Equipment class uses private variables with getters and setters.

## Abstraction

Database operations are hidden inside the EquipmentDAO class.

## Information Hiding

Database credentials are managed inside the DBConnection class.

## Association

Servlets communicate with the DAO class to perform CRUD operations.

## File Handling Implementation

File handling was implemented using Java FileWriter.

The system automatically records:

- Added equipment
- Updated equipment
- Deleted equipment

The log file:

gym-log.txt

Example:

Equipment Added: Treadmill  
Equipment Updated: Bench Press  
Equipment Deleted ID: 5

# User Interface Design

The system includes:

- Animated admin dashboard
- Professional dark UI
- Responsive cards
- Search bars
- Styled tables
- Hover animations
- Professional buttons
- Customer dashboard

The UI was designed using HTML and CSS.

## GitHub Contribution

GitHub was used for:

- Version control
- Branch management
- Individual contribution tracking
- Commit history management

## Challenges Faced

- JSP compilation errors
- Search functionality errors
- Session handling issues
- Broken count calculation problems
- GitHub branch management difficulties

These issues were solved through debugging and testing.

# Testing

The following functionalities were successfully tested:

| Function             | Status |
|----------------------|--------|
| Add Equipment        | Passed |
| Update Equipment     | Passed |
| Delete Equipment     | Passed |
| Search Equipment     | Passed |
| Login Authentication | Passed |
| File Handling        | Passed |
| Dashboard Counts     | Passed |

## Conclusion

The Gym Equipment Management System was successfully developed as a complete CRUD application using Java technologies and MySQL database.

The system fulfills all required CRUD functionalities while applying Object-Oriented Programming concepts, file handling, database connectivity, and professional UI design.

This project improved knowledge in:

- Java programming
- Database handling
- OOP concepts
- JSP and Servlets
- GitHub collaboration
- Frontend UI development

# References

- [Java Documentation](#)
- [MySQL Documentation](#)
- [Apache Tomcat Documentation](#)
- [JSP & Servlet Tutorials](#)
- [GitHub Documentation](#)